

Zachary HAMIDA, Ph.D.

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PROFESSIONAL EXPERIENCE

2023-present	Data Scientist Kruger Products Inc. - Advanced time series modelling for demand forecasting and pulp price prediction. - Deployed web apps/azure functions with UI functionality in Azure cloud platform. - Built and deployed RAG applications using Azure AI services and knowledge graph techniques. - Developed a multi-agent AI committee using CrewAI to automate the evaluation and ranking of grant applications at scale. - Developed and deployed a golden run application to help identify and highlight golden runs in manufacturing. - Identified key project requirements through meetings with end-users.
2023-2024	Data Science Consultant Beam Data Kruger Products Inc. - Worked on the development/enhancements of a demand forecasting model for consumer goods. - Participated in meetings with end-users and integrated their feedback into the forecasting model. - Integrated and deployed new customers accounts into the demand forecasting pipeline. - Enabled the use and testing of different forecasting models in the demand forecasting pipeline. - Developed a data science dashboard to enable model performance tracking. - Implemented machine learning operations MLOps standards to enhance decision-making on the forecasting model. - Proactive collaboration with team members to leverage diverse expertise and insights in solving technical problems.
2021-2023	Research Associate Polytechnique Montreal Transportation Ministry of the Quebec Province (MTQ) - Development of a reinforcement learning benchmark environment for emulating non-destructive evaluations on the bridge network in Quebec province. - Derived optimal decision-making methods tailored for maintenance planning. - Developed a standalone software for navigating and predicting the deterioration of infrastructure over time. - Led a team of entry-level researchers to identify problems in existing models and implement solutions using machine learning (ML) algorithms. - Monitor and maintain the performance of machine learning models.
2017-2020	Research Assistant Polytechnique Montreal Transportation Ministry of the Quebec Province (MTQ) - Designed and developed time-series models to forecast the deterioration based on visual inspections. - Conducted data analysis to identify patterns in non-destructive evaluation data from bridges. - Worked with MTQ to understand their decision-making requirements and translate them into data-driven solutions. - Contributed to the development of machine learning frameworks and libraries. - Provided technical guidance and mentoring to entry-level researchers.

TECHNICAL SKILLS

- Programming Languages: Python, MATLAB (App Designer, GPU compute), R
- Machine Learning Tools: PyTorch, Scikit-Learn, SKforecast, Ray, Streamlit, CrewAI
- Dev. Tools: AzureML, Azure DevOps, Docker, MLflow, TensorBoard, Github Actions, Azure Container Apps
- Applications: AI Agents, Graph RAG, Optimization, Forecasting Models, Deep Reinforcement Learning, Image Segmentation, Recommender Systems
- Cloud Computing Platforms: Azure AI, AWS

SOFTWARE DEVELOPMENT

Open-Source	InfraPlanner: (Python) Benchmark reinforcement learning environment for maintenance planning, - Emulate the deterioration process of transportation infrastructure over time. - Test and evaluate the effectiveness of maintenance policies. OpenIPDM: (MATLAB + App designer) Infrastructure probabilistic deterioration modelling, - Model the deterioration and effect of interventions over time. - Generate synthetic time series data and calibrate the deterioration model parameters. Azure DevOps AI Coder: (Python) Automated issue-to-PR pipeline using AI agents, - Plans, implements, tests, and reviews code changes from a GitHub issue. - Integrates with Azure DevOps pipelines and GitHub Actions for CI/CD.
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EDUCATION

2017-2020	Ph.D. ENGINEERING, Polytechnique Montreal , Montreal Prof. James-A GOULET (Adviser) Thesis: “Stochastic Modelling of Infrastructures Deterioration and Interventions based on Network-Scale Visual Inspections” Polytechnique Page
2014-2016	M.Sc. in COMPUTATIONAL SCIENCE, American University of Beirut , Beirut Prof. George SAAD (Adviser) & Prof. Fouad AZIZI (Co-Adviser) Thesis: “Hybrid Optimization Techniques for Oil Field Development”
2008-2013	B.Eng. in CIVIL ENGINEERING University of Aleppo , Aleppo Prof. Ammar KADAAN (Adviser) Final Year Project: “Structural Analysis & 4D Simulation for Multi-Storey Building”

CERTIFICATES & AWARDS

JAN. 2025	Candidat à la Profession d’Ingénieur. Ordre des ingénieurs du Québec.
SEP. 2023	AWS Certified Machine Learning - Specialty. Credential .
JUN. 2023	Machine Learning Engineering Certificate. Toronto Institute of Data Science and Technology.
APR. 2021	Higher Education Teaching Certificate. Harvard BOK Center.
DEC. 2016	AUB Startup Bootcamp by AltCity. 1st Place Award (Find A Nurse Startup Member).
SEP. 2014	Graduate Research Assistantship (GRA). American University of Beirut, Beirut, Lebanon.
SEP. 2014	Partial Scholarship in Visual Communication. Istituto Europeo di Design (IED), Florence, Italy.

MENTORING EXPERIENCE

Teaching	Teaching Assistant with Lecture Duties: - CIV6540: Probabilistic Machine learning for Civil Engineers (Winter, 2021 2023) - CIV8530: Structural and System Reliability (Fall, 2022)
Supervision	Research Project Mentoring: - Ali Fakhri, M.Eng. at Polytechnique Montreal (2021-2023, Degree earned). - Blanche Laurent, M.Eng. at Polytechnique Montreal (2020-2022, Degree earned).

ACADEMIC & SOCIAL SERVICE

Reviewer & Committee	- Journal of Geoenergy Science and Engineering - Developments in the Built Environment - International Symposium on Intelligent Technology for Future Transportation (ITFT) - International Probabilistic Workshop (IPW)
Moderator & Organizer	- Research Day of Structural Engineering Group at Polytechnique Montreal. - 9th Annual Postdoctoral Research/Career Day in Quebec Province.
Selection Committee	- NSERC scholarship. - Arbour Foundation scholarship.

JOURNAL PUBLICATIONS

1. Fakhri, S.A.K., **Hamida, Z.** & Goulet, J-A. (2025). "Scalable Probabilistic Deterioration Model based on Visual Inspections and Structural Attributes from Large Networks of Bridges". Journal of Advanced Engineering Informatics. [DOI](#)

2. **Hamida, Z.** & Goulet, J-A. (2024). "Quantifying the relative change in maintenance costs due to delayed maintenance actions on transportation infrastructure". *Journal of Performance of Constructed Facilities* DOI
3. Laurent, B., Deka, B., **Hamida, Z.** & Goulet, J-A. (2023). "Analytical Inference for the Inspectors Uncertainty based on Network-Scale Visual Inspections". *Journal of Computing in Civil Engineering*. DOI
4. **Hamida, Z.** & Goulet, J-A. (2023). "Hierarchical Reinforcement Learning for Transportation Infrastructure Maintenance Planning". *Reliability Engineering and System Safety*. DOI
5. **Hamida, Z.**, Laurent, B. & Goulet, J-A. (2022). "OpenIPDM: A Probabilistic Framework for Estimating the Deterioration and Effect of Interventions on Bridges". *SoftwareX*. DOI.
6. **Hamida, Z.** & Goulet, J-A. (2022). "A Stochastic Model for Estimating the Network-Scale Deterioration and Effect of Interventions on Bridges". *Struct. Control & Health Monitoring*. DOI.
7. **Hamida, Z.** & Goulet, J-A. (2021). "Quantifying the Effects of Interventions Based on Visual Inspections of Bridges Network". *Structure and Infrastructure Engineering*. DOI.
8. **Hamida, Z.** & Goulet, J-A. (2021). "Network-Scale Deterioration Modelling of Bridges Based on Visual Inspections and Structural Attributes". *Structural Safety*. DOI.
9. **Hamida, Z.** & Goulet, J-A. (2020). "Modeling Infrastructure Degradation from Visual Inspections Using Network-Scale State-Space Models". *Struct. Control & Health Monitoring*. DOI.
10. **Hamida, Z.**, Azizi, F. & Saad, G. (2017). "An Efficient Geometry-based Optimization Approach for Well Placement in Oil Fields". *Journal of Petroleum Science and Engineering*. DOI.

CONFERENCES, REPORTS & POSTERS

1. **Hamida, Z.** & Goulet, J-A. (2026). "OpenIPDM: Network-Scale Forecasting of Transportation Infrastructure Deterioration Using Global State-Space Models". 46th International Symposium On Forecasting | Montreal, Canada.
2. **Hamida, Z.** & Goulet, J-A. (2023). "Maintenance Planning for Bridges using Hierarchical Reinforcement Learning". 14th International Conference on Applications of Statistics and Probability in Civil Engineering. Dublin, Ireland.
3. Fakhri, S.A.K., **Hamida, Z.** & Goulet, J-A. (2023). "Bayesian neural networks for large-scale infrastructure deterioration models". 14th International Conference on Applications of Statistics and Probability in Civil Engineering. Dublin, Ireland.
4. **Hamida, Z.** & Goulet, J-A. (2023). "OpenIPDM: Une librairie ouverte pour modéliser la dégradation d'un parc d'infrastructures". 29e Colloque sur la progression de la recherche québécoise concernant les ouvrages d'art. Québec, Canada.
5. **Hamida, Z.** & Goulet, J-A. (2022). "Modelling the Deterioration of Infrastructures Using Network-Scale Visual Inspections". 11th International Conference on Structural Health Monitoring of Intelligent Infrastructure. Montreal, Canada.
6. Laurent, B., **Hamida, Z.** & Goulet, J-A. (2022). "Estimating the Bias Associated with Inspectors in the Context of Visual Inspections on Infrastructures". 11th International Conference on Structural Health Monitoring of Intelligent Infrastructure. Montreal, Canada.
7. **Hamida, Z.** & Goulet, J-A. (2021). "Prédire la dégradation et comprendre l'effet des interventions : une méthode d'apprentissage machine adaptée aux rapports d'inspection issus d'une large population de structures". Technical Report for the Ministry of Transport in Quebec (MTQ).
8. **Hamida, Z.** & Goulet, J-A. (2019). "State-Space Models for Network-Scale Analysis of Bridge Inspection Data". 13th International Conference on Applications of Statistics and Probability in Civil Engineering. Seoul, South Korea.
9. **Hamida, Z.** & Goulet, J-A. (2019). "Modeling Infrastructure Degradation from Visual Inspections Using Network-Scale State-Space Models". Modeling and Numerical Methods for Uncertainty Quantification (MNMUQ 2019), Porquerolles Island, France.

ACTIVITIES

- Cycling, Stand up PaddleBoard, Running, Basketball & Guitar.

REFERENCES

- Available upon request.